New power series from old — 6.2

Example: We know $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ on $(-1,1)$, so if we want

$$\frac{1}{2-x} = \frac{1}{2}\cdot\frac{1}{1-\frac{x}{2}} = \frac{1}{2}\sum_{n=0}^{\infty}\left(\frac{x}{2}\right)^n = \sum_{n=0}^{\infty}\frac{x^n}{2^{n+1}}$$

holds for x with $\frac{x}{2}\in(-1,1)$, i.e. for $x\in(-2,2)$.

Similarly,

$$\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty}(-x^2)^n = \sum_{n=0}^{\infty}(-1)^n x^{2n}$$

holds for x with $-x^2\in(-1,1)$, i.e. for $x\in(-1,1)$.

How to re-center?

$$\frac{1}{1-x} = \frac{1}{2-(x+1)} = \sum_{n=0}^{\infty}\frac{(x+1)^n}{2^{n+1}}$$

holds for x with $x+1\in(-2,2)$, i.e. for $x\in(-3,1)$.

Same function, centered at -1.

Theorem: If $\sum c_n (x-a)^n$ has radius of convergence $R > 0$, then $f(x) = \sum c_n (x-a)^n$ is differentiable on $(a-R, a+R)$ and we can find $f'(x)$ as a power series by differentiating term-by-term.

That is, $f'(x) = \sum n c_n (x-a)^{n-1}$ with the same radius of convergence.

Similarly, $\int f(x)\,dx = C + \sum \frac{c_n}{n+1} (x-a)^{n+1}$ again with radius of convergence R.

<u>Example</u>: $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ means we can find

a power series for $\frac{d}{dx}\frac{1}{1-x} = \frac{-1}{(1-x)^2}\cdot -1 = \frac{1}{(1-x)^2}$

by differentiating each term of the power

series $\frac{d}{dx}\sum_{n=0}^{\infty} x^n = \sum_{n=1}^{\infty} n x^{n-1} = \sum_{n=0}^{\infty} (n+1)\, x^n$

Conclusion $\frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} (n+1)\, x^n$ with

radius of convergence $R = 1$.

(Interval of convergence here is $(-1,1)$ still.)

Example: Know that $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-x)^n = \sum_{n=0}^{\infty} (-1)^n x^n$.

Integrating gives $\ln|1+x| = C + \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} x^{n+1}$

with radius of convergence 1.

Plugging in $x=0$ gives $\ln(1) = C + \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} (0)^{n+1}$

$\iff C = 0$, that is

$$\ln(x+1) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} x^{n+1} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

(no absolute value bars since we work on $(-1,1)$ where $x+1>0$ anyway)

$x=-1$: $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} (-1)^n = -\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

$x=1$: $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ which converges by the AST.

$$\ln(x+1) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

with interval of convergence $(-1, 1]$.

Tells us also that $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \ln(1+1) = \ln(2)$.

Know also $\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$ w/ RoC $= 1$

Thus $\arctan(x) = \int \frac{1}{1+x^2}\,dx = C + \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$

$0 = \arctan(0) = C + \sum 0 = C,$

$\arctan(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$

$x=-1$: $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} (-1)^{2n+1} = \sum_{n=0}^{\infty} \frac{(-1)^{3n+1}}{2n+1}$ converges by the AST

$x=1$: $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$ converges by the AST.

So $\arctan(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$ on $[-1, 1]$.